

$$\begin{array}{l}
 \text{Row 3} = \text{Row 3} \\
 \times 1/16 \rightarrow \begin{bmatrix} 1 & 2 & 7 \\ 0 & 9 & 18 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{\text{Row 2} = \text{Row 2} \times 1/9} \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{\text{Row 3} = \text{Row 3} - \text{Row 2}} \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \\
 \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{Row 1} = \text{Row 1} - 2 \times \text{Row 2}} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}
 \end{array}$$

The solution of (3) is $x_1 = 3$ and $x_2 = 2$. Hence b is a linear combination of a_1 and a_2 , with weights $x_1 = 3$ and $x_2 = 2$. That is,

$$3 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix} \quad \square$$

Observe in Example 5 that the original vectors a_1, a_2 and b are the columns of the augmented matrix that we row reduced:

$$\begin{bmatrix} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{bmatrix}$$

For brevity, write this matrix in a way that identifies its columns—namely,

$$[a_1 \ a_2 \ b] \quad (4)$$

Take the vectors in the order in which they appear in (1) and put them into the columns of a matrix as in (4).

Fundamental Fact

A vector equation $x_1 a_1 + x_2 a_2 + \dots + x_n a_n = b$ has the same solution set as the linear system whose augmented matrix is $[a_1 \ a_2 \ \dots \ a_n \ b]$. In particular, b can be generated by a linear combination of $a_1, \dots, a_n$ if and only if there exists a solution to the linear system corresponding to the matrix (5).

GV. Imp.

One of the key ideas in linear algebra is to study the set of all vectors that can be generated or written as a linear combination of a fixed set $\{v_1, v_2, \dots, v_p\}$ of vectors.